ADR-0003: Simulated contestant behavior and the tiebreaker rule

Status: accepted. Supersedes ADR-0002. Date: 2026-10-13

Context

Two defects in the Tier 2 and Daily Double engines surfaced on 2026-10-13.

1. **Wrong tie rule.** The engines counted a tie for first as a full win, on the grounds that co-champions both return. That rule ended with the 24 November 2014 episode. Since then, a tie for first after Final Jeopardy is settled by a tiebreaker clue, and only one player returns (The Ringer, 2021). The visible symptom: for a leader whose cover bet was $6,001, Smart Bet recommended $6,000 — a bet that only ties second place if they double up. Under the real rule that tie is worth about half a win.
2. **Placeholder opponents.** ADR-0002 shipped a uniform split over each opponent's own Tier 1 breakpoints, at 50% accuracy, independent across players. It was labeled as a placeholder. Tesauro et al. (2013, JAIR 47:205-251) publish enough about real contestant behavior to do better.

Decision

All values live in one configuration block in `engine/behavior.py`, mirrored in the `BEHAVIOR` object in `app/index.html`.

Scoring. A player's result is a *win share*: 1 for an outright win, 1/k for a k-way tie for first (equal tiebreaker odds, assumed), 0 otherwise. Nobody wins if the top score is $0 or less.

Opponent wagers. Each opponent is assigned a role by score: leader (A), second (B), third (C).

Role	Bet types and weights	Source
B	all-in 26%, keep C out (bet < B−2C, or cover 2C−B+1) 27%, overtake A (A−B+1) 15%, two-thirds (3B−2A−1) 8%, random 24%	PUBLISHED for the regime B ≥ ¾A and B ≥ 2C (T13 §2.3). ASSUMED for the other five regimes.
A, not locked	cover 2B−A+1 75%, tie-cover 2B−A 10%, all-in 5%, random 10%	ASSUMED. T13 §1.1 says A "usually" covers, "frequently" 2B−A+1, with no split.
A, lockout (A > 2B)	$0 50%, largest safe bet A−2B−1 50%	ASSUMED.
C	random	PUBLISHED that C's bets do not depend on the game state (T13 §2.3 note 5). The flat shape is ASSUMED.

"Random" is an equal spread over 0, ¼, ½, ¾ and all of the score. When a bet type does not exist in a state (for example, no third player to keep out), its weight is shared out in proportion among the others.

Accuracy. Opponents answer Final Jeopardy right 50% of the time (PUBLISHED, average contestants, T13 §2.3; champions 60%). Right/wrong outcomes are correlated across all players at 0.3 (PUBLISHED, T13 §2.3), modeled as a one-factor Gaussian copula. We read 0.3 as the binary correlation at 50% accuracy and convert it with Sheppard's formula to a latent correlation $r = \sin(0.3\pi/2) \approx 0.454$. The user is inside the same copula: a hard clue is hard for everyone.

Daily Double. Future Daily Doubles are answered right 64% of the time (PUBLISHED, average contestants, T13 §2.2). Inside a rollout, opponents' Final Jeopardy wagers are drawn from the model above.

Computation. Tier 2 is exact: it integrates over the shared factor (trapezoid rule, step 0.25 on $[-6, 6]$) and enumerates every wager and right/wrong branch. The DD engine samples the same model.

Consequences

Positive:

- Smart Bet no longer recommends a tie-only bet when one more dollar wins.
- Every opponent number is either cited or labeled as an assumption, in one place.
- Correlated outcomes capture a real effect: when you miss, a rival is more likely to have missed too.

Negative:

- Five of the six B regimes, all of A's split and C's shape are our assumptions, not fitted. The published B weights came from groups of "only a few hundred samples" each (T13 §2.3), so even those carry wide error bars.
- The data are from roughly the mid-1990s to 2012. Contestant behavior has likely changed since; non-stationarity is not modeled.
- Equal tiebreaker odds is an assumption; buzzer skill decides the clue.
- The reading of "$\rho \approx 0.3$" as a binary correlation is our interpretation; the paper does not say which scale it means.
- Win percentages from any earlier version are not comparable: they scored ties as wins against a different opponent model.
- The app's Daily Double calculation takes about twice as long, because each rollout now simulates opponent wagers.